

NOTES ON BAUR'S LECTURES AND PAPER

DERRICKSON YMBON

Department of Mathematics, North-Eastern Hill University, Shillong, India

Abstract: These are the notes from a series of lectures on the geometric model of the module category of gentle algebras that Prof. Karin Baur gave at the CIMPA School 2026 held in Thanjavur, India in July 2026. The notes consist of a mix of plain lectures and tutorial sessions conducted by Prof. Baur herself. They first start with the basic concepts and definitions of gentle and string algebras. They then go on to describe partial triangulations and the corresponding tiling algebras, and after that we go to strings and string modules, which are the central objects of study here. Finally a geometric module to describe the module category for the tiling algebras is developed.

Acknowledgements: These notes are based primarily on the lectures delivered by Prof. Karin Baur as well as her paper [1]. I am very grateful to Prof. Baur for the lectures, explanations, and insights that provided the foundation for these notes. I am also grateful to CIMPA for letting me attend the school. It has provided an immense amount of knowledge and experience, especially since it is the first CIMPA School I have attended

The material has been written and organized by me based on my understanding of the lectures. I have occasionally added further explanations, examples, or references where I found them helpful for presenting the material more clearly. The solutions to the problems posed by Prof. Baur have been presented in these notes. Sources consulted beyond the lectures are cited where appropriate.

Any errors, omissions, or inaccuracies in these notes are my own.

1. GENTLE/LOCALLY GENTLE ALGEBRAS, STRINGS AND STRING ALGEBRAS

Definition 1.1. Let k be an algebraically closed field, let $Q = (Q_0, Q_1, s, t)$ be a quiver and let $A = kQ/I$ denote its path algebra. Then a finite dimensional algebra is said to be *gentle* if it is of the form kQ/I , where (Q, I) satisfies the following conditions:

- (1) The in-degree and out-degree of every vertex of Q is at most 2
- (2) For every arrow a of Q , there exists at most one arrow b with $s(b) = t(a)$ and at most one arrow c with $s(a) = t(c)$ such that $ab \in I$ and $ca \in I$
- (3) For every arrow a of Q , there exists at most one arrow b with $s(b) = t(a)$ and at most one arrow c with $s(a) = t(c)$ such that $ab \notin I$ and $ca \notin I$
- (4) The ideal I is admissible (i.e. $I \subset R^2$ where R^2 is the arrow ideal) and is generated by paths of length 2.

The algebras kQ/I that satisfy conditions 1,3 and 4 (however I need not be admissible) are called *string* algebras. *Locally* gentle algebras are the algebras which satisfy all of the conditions above but can have infinite dimension as well.

Example 1.2. (1) Consider the following quiver Q :

$$\begin{array}{ccccc} 1 & \xrightarrow{a} & 2 & \xrightarrow{c} & 3 \\ & \xrightarrow{b} & & & \end{array}$$

Here, we consider $I = \langle ac \rangle$ The algebra kQ/I for the above quiver is gentle as well as string as well as locally gentle

- (2) The same quiver but with a different ideal gives a different algebra:

$$\begin{array}{ccccc} 1 & \xrightarrow{a} & 2 & \xrightarrow{c} & 3 \\ & \xrightarrow{b} & & & \end{array}$$

Here , we consider $I = \langle ac, bc \rangle$. The algebra we get from the above quiver is string but not gentle.

(3)

$$1 \begin{array}{c} \xrightarrow{a} \\ \xrightarrow{b} \end{array} 2 \xrightarrow{c} 3 \xrightarrow{d} 4$$

Here, we consider $I = \langle ac, bcd \rangle$. Again the corresponding algebra is a string algebra which is not gentle.

Thus we can see that string algebras are a class of algebras that contain gentle algebras, and we can see that there are string algebras that are not gentle from the above examples.

2. TILING ALGEBRAS

Here, we will describe algebras obtained from partial triangulations of marked oriented surfaces.

Let (S, M) denote a marked oriented surface. An *arc* γ in (S, M) is a curve in S which satisfies the following conditions:

- *₁ The endpoints of γ are elements of M
- *₂ γ intersects ∂S only at its endpoints
- *₃ γ does not cut out a monogon or a digon
- *₄ γ does not intersect itself, except possibly at its endpoints

All the arcs are defined upto homotopy. A *partial triangulation* P is a collection of pairwise non-crossing arcs in the surface.

The following are examples of partial triangulations of surfaces

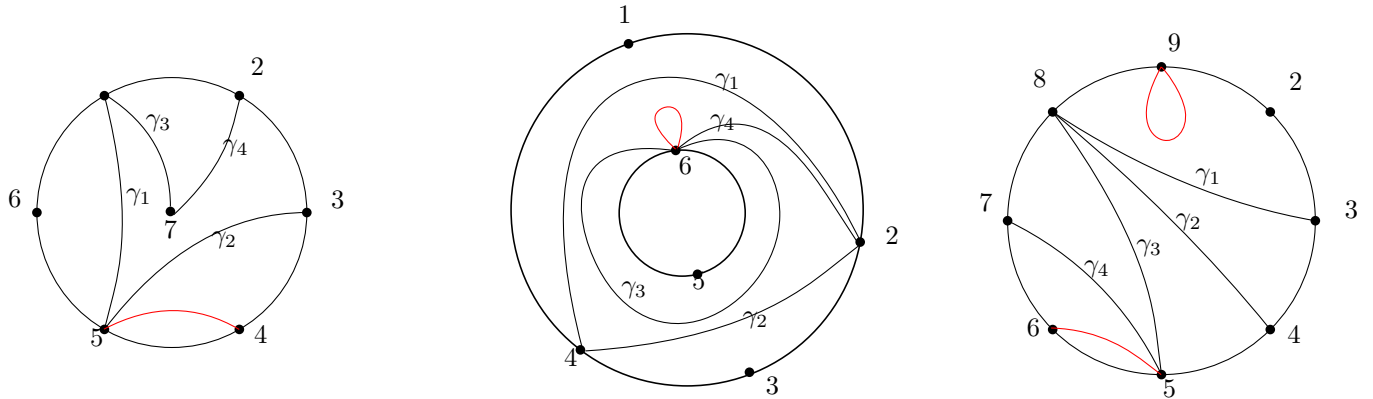


FIGURE 1. Partial triangulations

The curves drawn in red are the curves which violate the conditions *_•.

The triple (S, M, P) will be referred to as a tiling, and from (S, M, P) , we can obtain a quiver Q_P as follows:

Definition 2.1. The vertices of Q_P are in bijection with the elements of P . That is, $(Q_P)_0$ consists of the arcs in P , and the arrows are given by clockwise rotations around a common endpoint of any two arcs, provided there are no arcs in between. We can talk about clockwise rotations from the fact that we use an oriented surface.

The quivers for the partial triangulations in the above examples are the following

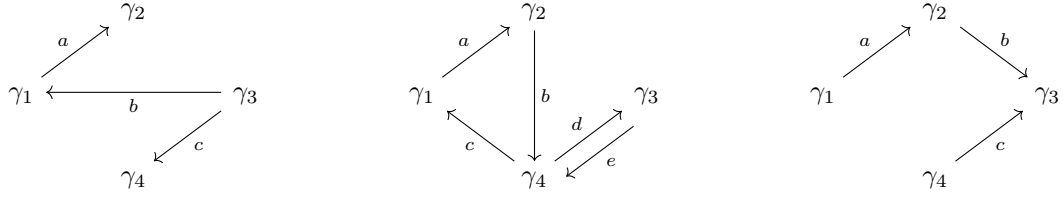
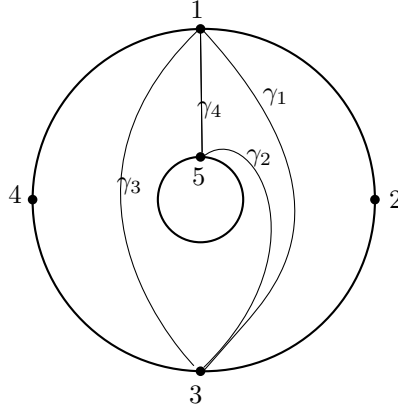


FIGURE 2. Quivers from the tilings above

A triangulation of (S, M) is a partial triangulation which dissects the surface into triangles. In other words, a triangulation T is a maximal collection of arcs which do not intersect in the interior of the surface.

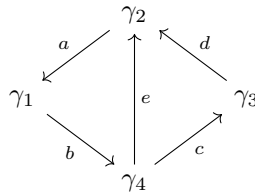
3. TUTORIAL SESSION 1

Question 3.1. Consider the following tiling (S, M, P)



- (1) Find Q_P, I_P
- (2) Complete P to triangulations P_1 and P_2

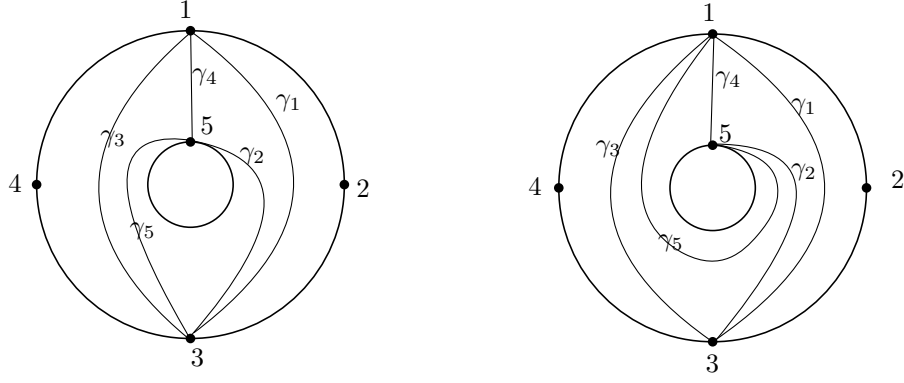
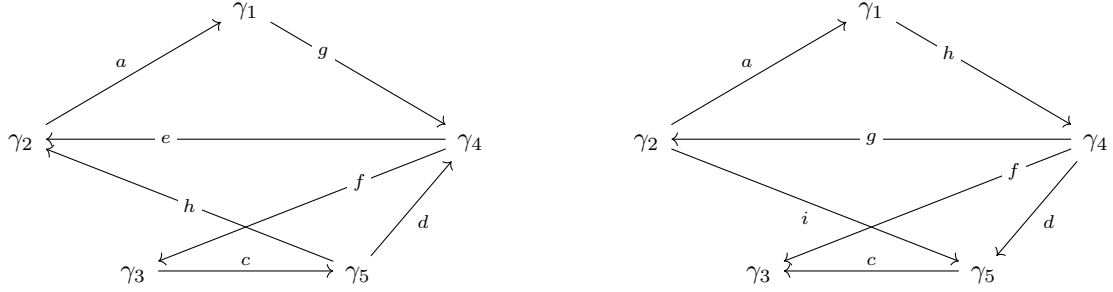
Solution 3.1 The quiver Q_P is given by



and I_P , as defined below (see Definition 3.3), is generated by all the paths ab where a, b are two arrows inside the same tile which can be composed. So we get that $I_P = \langle ab, be, ea \rangle$.

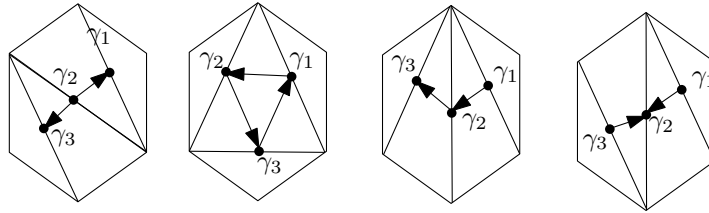
The triangulation given by P can be completed to triangulations P_1 and P_2 as follows:

and their corresponding quivers Q_{P_1} and Q_{P_2} are

FIGURE 3. P_1 and P_2 

Question 3.2. Find all quivers coming from triangulations of a hexagon

Solution 3.2 There are 14 possible triangulations of a hexagon, of which the possible quivers are the following



Definition 3.3. The tiling algebra $A_P = kQ_P/I_P$ corresponding to the partial triangulation P is the algebra formed where the quiver Q_P is the quiver as described in 2.1 and the ideal I_P is generated by paths ab of length 2 which satisfy the one of the following conditions :

- (1) Either $a = b$, that is the arc $\mathbf{a}$ is a loop, or
- (2) $a \neq b$, and either $p_a \neq p_b$, or we have that $p_a = p_b$ and thus we have the following cases

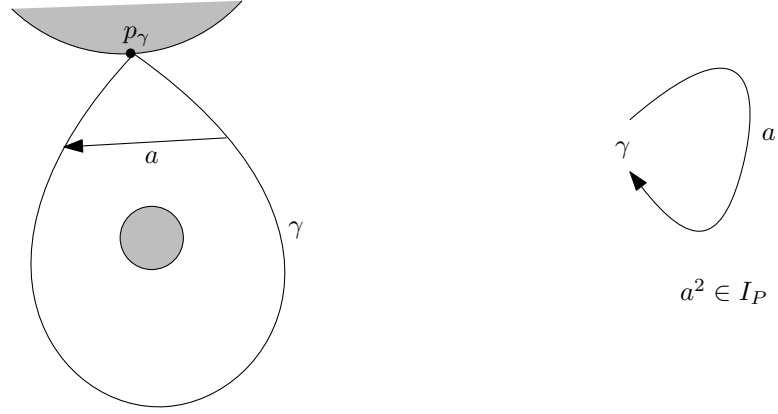
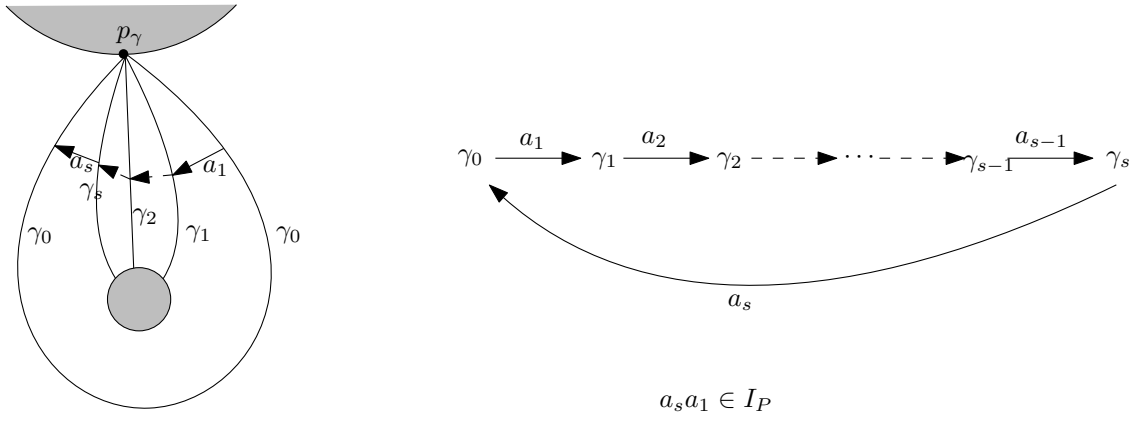

 FIGURE 4. Case 1: γ is a loop


FIGURE 5. Case 2

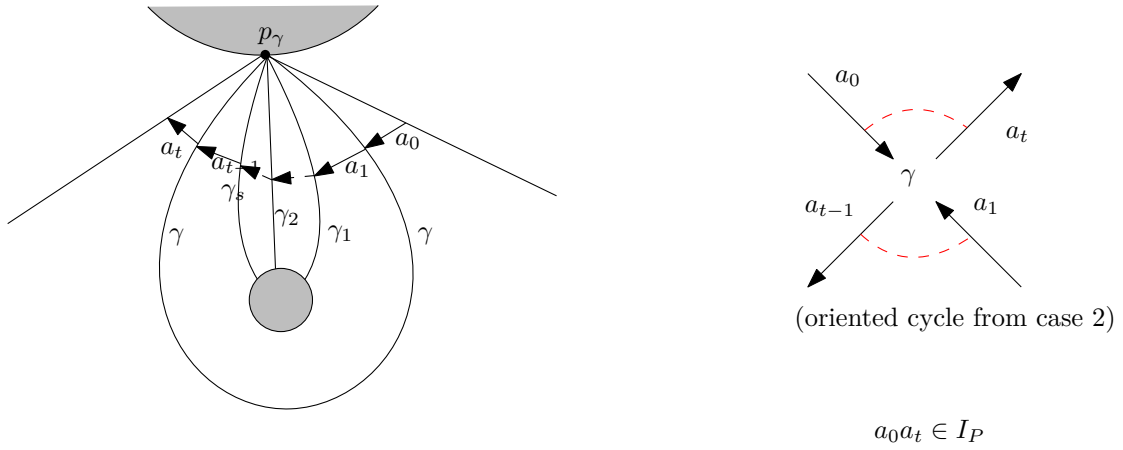


FIGURE 6. Case 3

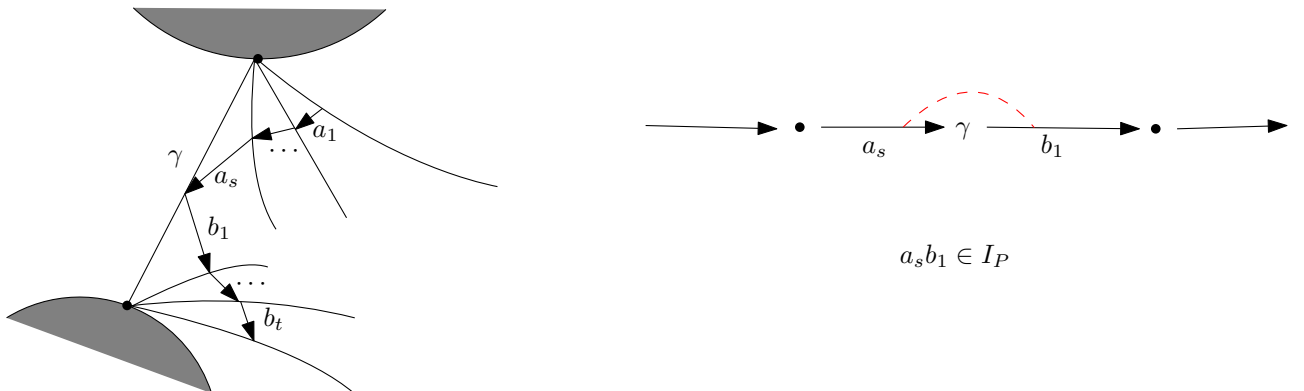


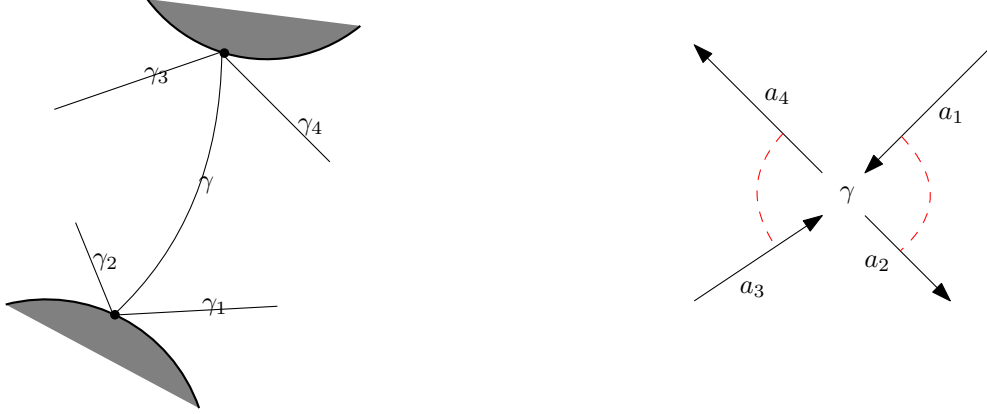
FIGURE 7. Case 4

If an algebra A_P can be described like this, we call it a tiling algebra.

Remarks 3.4. (1) If $a, b \in (Q_P)_1$ with $t(a) = s(b)$ are such that $ab \in I_P$, then the arcs corresponding to $s(a), t(a) = s(b)$ and $t(b)$ bound the same region in (S, M, P) .
 (2) If P is a triangulation of (S, M) with no punctures, then A_P is a Jacobian algebra. In that case any oriented 3-cycle is full of relations

Theorem 3.5. An algebra A is gentle $\iff A$ is a tiling algebra, i.e. $A \cong A_P$ for some (S, M, P) , where S does not contain any punctures.

Outline of Proof: ($\Leftarrow$) Consider the following diagram, which is a maximal configuration of a triangulation for a quiver:



It is easy to see from the above figure that since the surface and the resulting quiver is a maximal configuration, that is, it is a configuration where there are as many arcs as possible and all the types of tiles are accounted for, which results in the above quiver, which again is a maximal configuration of a gentle algebra. Therefore, it is easy to see that the conditions for a gentle algebra are satisfied:

- (1) $I_P \subset R^2$ by definition
- (2) If γ is an arbitrary vertex in the quiver, again since this is a maximal configuration, we get that there are at most two incoming arrows and two outgoing arrows at γ
- (3) Here, we get that $a_1 a_2 \in I_P$, $a_3 a_4 \in I_P$, so the third condition for a gentle algebra is satisfied for every arrow
- (4) Similarly here, we get that $a_3 a_2, a_1 a_4 \notin I_P$ so again, the fourth condition is satisfied for a gentle algebra. $\square$

The proof for the reverse direction is given in [4][Section 1.4, Keywords: Γ_A , lamination]

From here on, we assume that in (S, M, P) , all the tiles formed by the arcs in P which satisfy one the following conditions :

- (1) The tiles are triangles with 2 boundary segments as the sides
- (2) The tiles are n -gons with $m \geq 3$, which has at most one boundary segment as one of its sides.
- (3) The tiles look like the following. They are 1-gons whose interior contains exactly one unmarked boundary component. These tiles will be called **Type-I** tiles

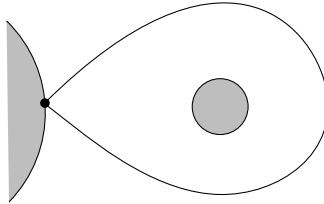


FIGURE 8. Type I tile

- (4) The tiles look like the following. They are 2-gons, whose interior contains exactly one unmarked boundary component. These tiles will be called **type II** tiles

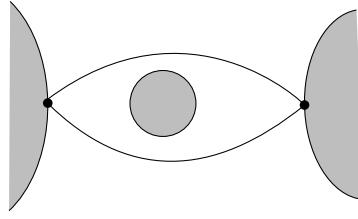
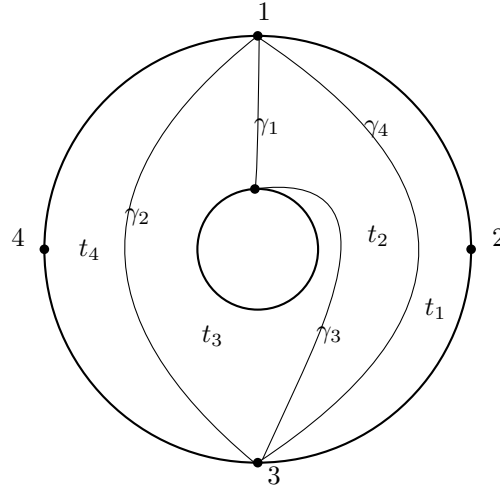


FIGURE 9. Type 2 tile

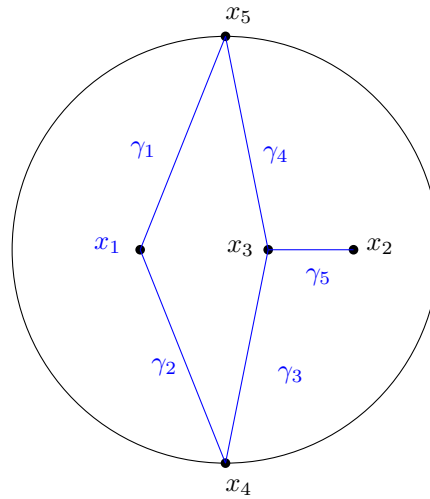
For example, let us take the following figure and let us label the tiles as follows



Here, the tiles t_1 and t_4 satisfy condition 1, while tiles t_2 and t_3 satisfy condition 2)

4. TUTORIAL 2

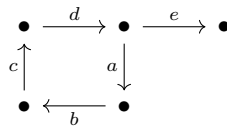
Consider the following tiling (S, M, P)



Question 4.1. Find Q_P, I_P

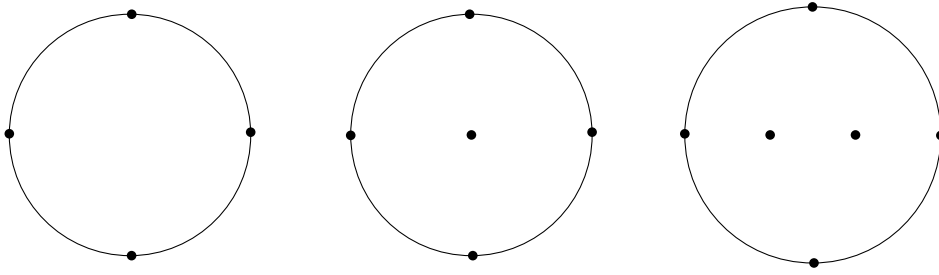
Question 4.2. For the 4 quivers of the triangulations of a hexagon, work out I in each case, and find a basis for each kQ/I

Question 4.3. Consider the following quiver



with $I = \langle da, ab, cd \rangle$. Find a basis of kQ/I

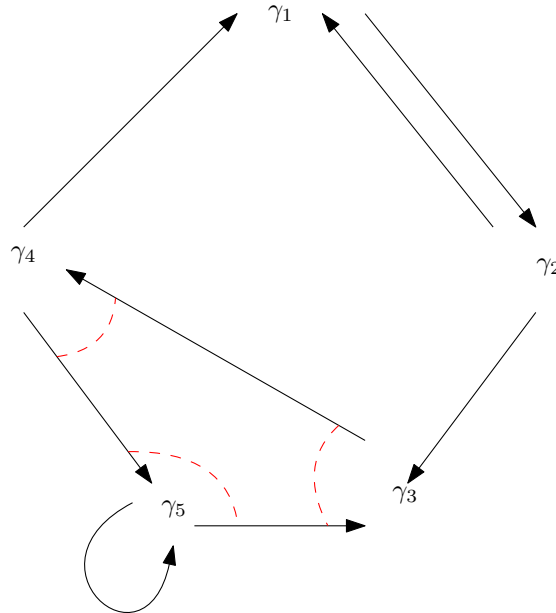
Question 4.4. Consider the following surfaces with marked points



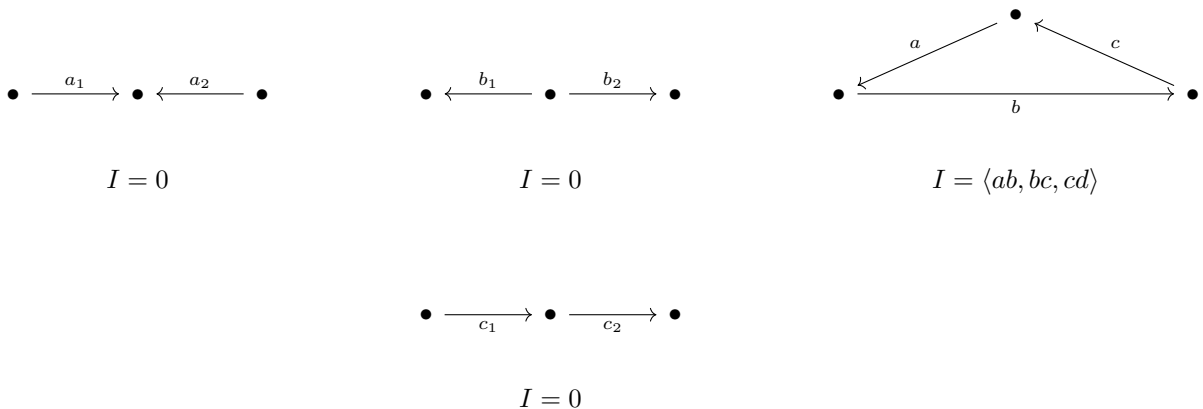
Find a triangulation for each marked surface and work out the associated quiver. Can you find a rule for how the number of arcs in a triangulation changes for each marked point that is added?

Question 4.5 (No of arcs in any triangulation T of (S, M)). Assume (S, M) has does not have empty boundaries, meaning that the boundary will have at least one marked point. Then $n = \text{rank of } (S, M, T) = \text{no of arcs in } T = 6g + 3b + 3p + c - 6$, where b is the number of connected components of ∂S , p is the number of punctures and $c = |M| - p$, i.e, c is the number of marked points in the boundary

Solution 4.1. The quiver Q_P as well as I_P is the following



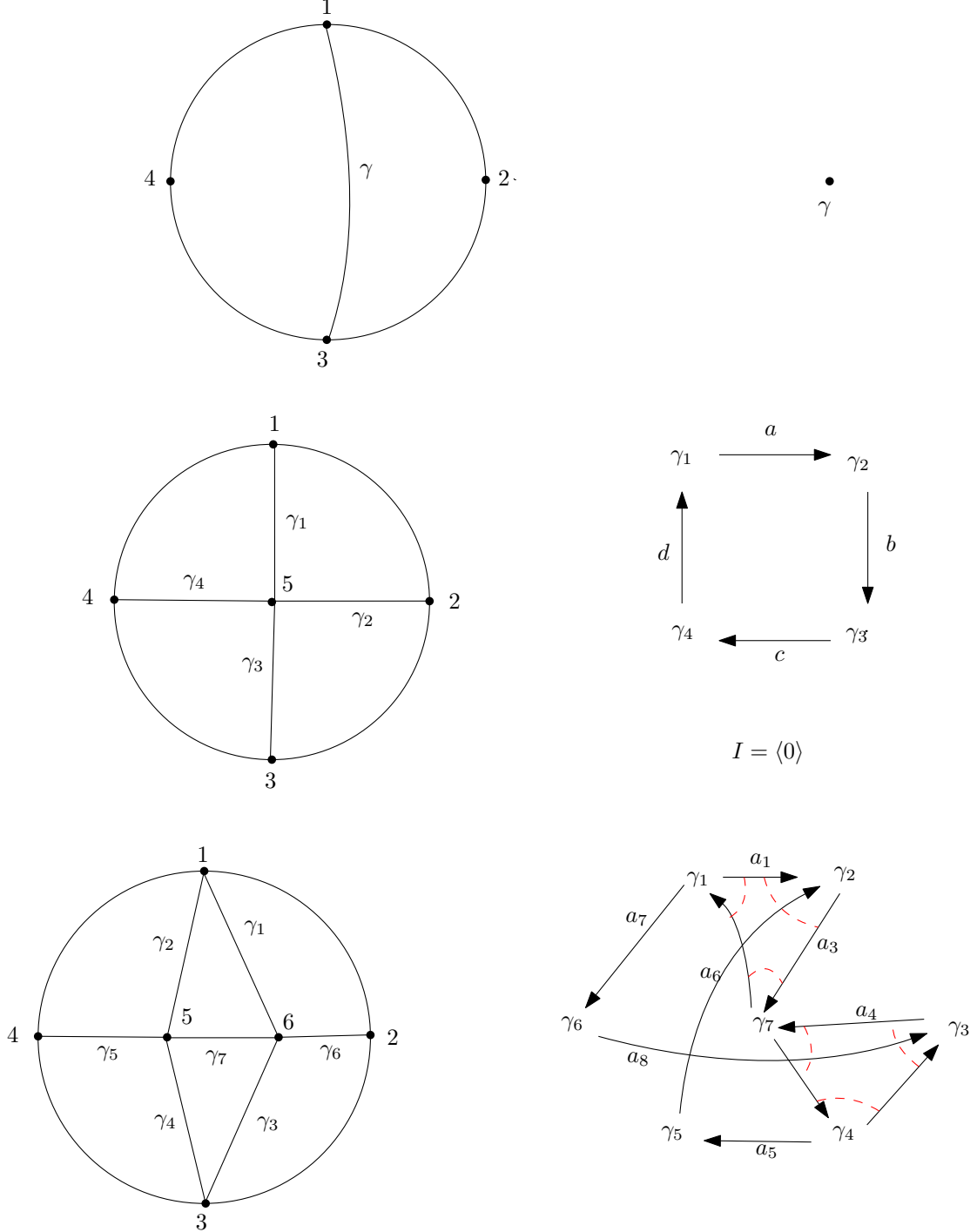
Solution 4.2 The quivers for the triangulations of a hexagon are the following



Then the bases for each of the algebras are $\{e_1, e_2, e_3, a_1, a_2\}$, $\{e_1, e_2, e_3, b_1, b_2\}$, $\{e_1, e_2, e_3, a, b, c\}$ and $\{e_1, e_2, e_3, c_1, c_2, c_1c_2\}$. Here, while the vertices are not labelled, the e'_i 's are the trivial paths at each vertex i

Solution 4.3. A basis of kQ/I is the set $\{e_1, e_2, e_3, a, b, c, d, e, bc, de\}$

Solution 4.4. Triangulations for each marked surface (S, M) along with the quivers and ideal are as follows



5. STRINGS AND BANDS

Now we state the definition of strings. First, we denote the opposite quiver of a quiver Q by Q^{op} , which is nothing but the quiver with the same set of vertices but with the arrows all inverted. That is, if a is an arrow in Q with $s(a) = i$ and $t(a) = j$, then in Q^{op} , denoting the inverse of a by a^{-1} , $s(a) = j$ and $t(a) = i$.

Definition 5.1. A string w in a quiver Q is a walk inside $Q \cup Q^{op}$ such that w does not contain any subpath which is a relation and such that no arrow is followed by its inverse. That is, w does not contain any 2-cycle of the form aa^{-1} and if ab is a relation in Q or Q^{op} then ab is not a subword of w

In example 1.2(1), $w = c^{-1}a^{-1}b$ is a string, ab^{-1} is a string and so on
Another example:

$$1 \xrightarrow[a]{a} 2 \xrightarrow{c} 3$$

where $I = \langle ac \rangle$. Here $w = c^{-1}b^{-1}a$ is a string but $c^{-1}a^{-1}b$ is not a string since $ac \in I$

Definition 5.2. A band b is a string $a_1 \cdots a_n$ such that $s(a_1) = t(a_n)$, b^n is a string for any $n > 0$ and b itself is not the power of any substring.

To every string w , we can associate a string module $M(w)$, where we place a copy of the field k each time the string w goes through a vertex v in Q , and the action of an arrow $a \in Q$ is given by the identity maps if $a \in w$ and 0 elsewhere.

For the string $w = c^{-1}a^{-1}b$, we have the following string module:

$$k^2 \xrightarrow{\begin{bmatrix} 10 \\ 01 \end{bmatrix}} k^2 \xrightarrow{\begin{bmatrix} 10 \end{bmatrix}} k$$

Note: Two string modules $M(w)$ and $M(v)$ are isomorphic if and only if $v \in \{w, w^{-1}\}$. Similarly to every band b we associate a family of band modules $M(b, V, \varphi) \in \text{mod } A$ where V is a finite-dimensional k -vector space, and φ is an irreducible automorphism.

If $k = \bar{k}$, then the band modules are given by $M(b, n, \varphi)$, where n is the dimension of V and

$$\varphi = J_n(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}. \text{ In a band module, we place a copy of the field } k^n \text{ at every}$$

vertex the band passes through, and the action of an arrow a in Q in $M(b, n, \varphi)$ is given by the identity morphisms if $a = a_i$ for $i = 1, 2, \dots, n-1$, and φ if $a = a_n$.

6. TUTORIAL 3

Question 6.1. Let Q be the following quiver

$$1 \xleftarrow[e]{e} 2$$

Here the non-zero strings are $\{1, 2, e, f, fe^{-1} \cong ef^{-1}, e^{-1}f \cong f^{-1}e, f^{-1}ef, \dots\}$. Find the string modules for the given strings. Also find the projectives and injectives of the above quiver

Find a band and write the corresponding band module upto $n = 3$

Soln: The string modules in order of the strings given in the question are as follows

$$\begin{aligned} k &\xleftarrow[0]{0} 0 \\ 0 &\xleftarrow[0]{0} k \quad k \xleftarrow[0]{1} k \\ 0 &\xleftarrow[1]{0} k \\ k^2 &\xleftarrow{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} k \end{aligned}$$

One band is the string $b = e^{-1}f$

Then the band module $M(b, 1, \varphi)$ is the representation
and the band module $M(b, 2, \varphi)$ is the module

$$k \begin{array}{c} \xleftarrow{\lambda_\varphi} \\ \xleftarrow{id} \end{array} k$$

$$k^2 \begin{array}{c} \xleftarrow{id} \\ \left(\begin{array}{c} \lambda 1 \\ 0 \lambda \end{array} \right) \end{array} k^2$$

7. GEOMETRIC MODEL:

Let $(S, M, P) := S$ be a partial triangulation of a marked oriented surface. Let A be the gentle algebra given by this partial triangulation. The goal here is to get full description of the string/module in terms of arcs in the surface. In other words, there is a bijection between the set of arcs γ not in P and the set of indecomposable A -modules $M(\gamma)$, where A is the tiling algebra obtained from (S, M, P) . This arc will have to satisfy the following properties:

- (1) If γ_1 and γ_2 are homotopic, then $M(\gamma_1) = M(\gamma_2)$.
- (2) If γ does not intersect any other arc in P , then $M(\gamma) = 0$
- (3) If γ intersects P but goes through a relation, then $M(\gamma) = 0$

Remark: If P is not a triangulation, then we can get γ_1, γ_2 not homotopic such that $M(\gamma_1) \neq M(\gamma_2)$.

In other words, if γ is an arc not contained in P , then when P gives the quiver of the algebra, then the intersection of γ with P gives the support of the indecomposable A -module. An example illustrating that non homotopic arcs can describe the same module is in the following figure

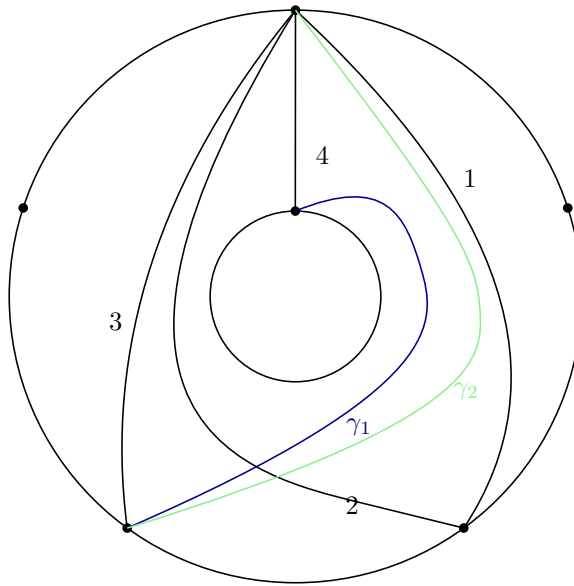
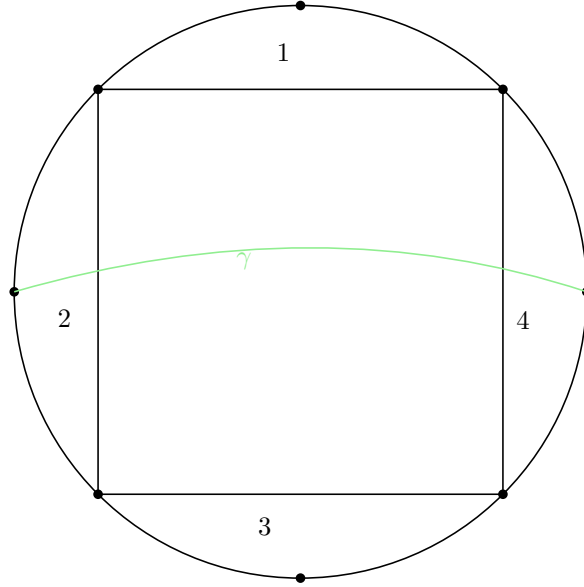
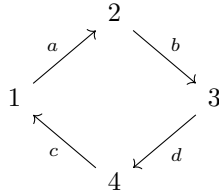


FIGURE 10. $M(\gamma_1) = M(\gamma_2)$

For an example of the 3rd property consider the following (S, M, P)



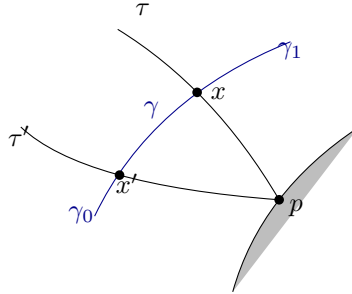
The quiver for the above partial triangulation is the following



with relations everywhere. But there is no indecomposable module with support only at 1 and 3, since if there was, it is always the direct sum of the simple modules at 1 and 3

Definition 7.1. Let $(S, M, P) = S$ be a surface. Then an arc γ not in P is called a *permissible* arc if it satisfies the following conditions:

- a) If γ_0, γ_1 go around an empty boundary component and the winding number is at most 1. Here γ_0 and γ_1 are the endsegments of γ with the first and last crossing respectively with P , and the winding number is the number of times γ_i travels around the empty boundary component, for $i = 0, 1$
- b) If γ crosses $\tau, \tau' \in P$ consecutively, meaning that there are no other arcs in between τ and τ' that γ intersects with, then τ, τ' meet at a point $p \in M$ and form a triangle with γ near p , as in the following figure



Definition 7.2. A *permissible closed curve* is simply a closed curve that satisfies 1(b)

Remarks 7.3. (1) Boundary segments and boundary arcs of P are permissible, since if α is a boundary arc, that is, it is an arc that is homotopic to another curve δ on the boundary that intersects M only at the endpoints, then take $\tau = \tau' = \delta$ to be δ itself. Then clearly the winding number is 0, and it forms the.

- (2) Any two consecutive crossings of γ with P correspond to an arrow of $A = A_P$
- (3) If P is a triangulation of the surface, then every arc is permissible.

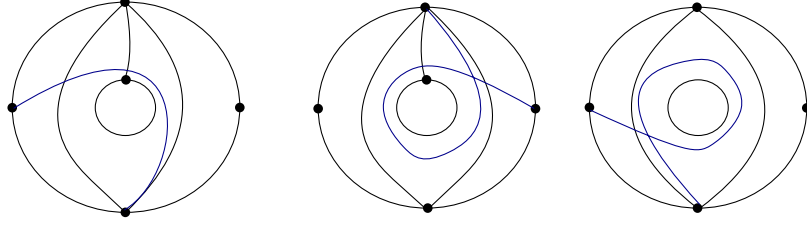


FIGURE 11. Permissible arcs

The following are examples of permissible arcs

While the following are non examples, where condition 1(b) is violated

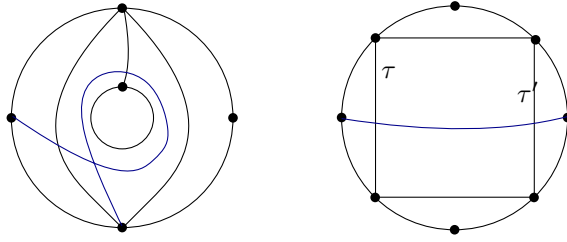
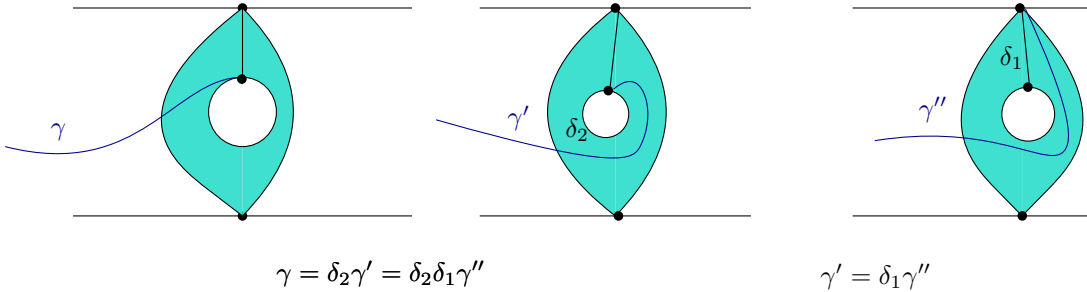


FIGURE 12. Non-permissible arcs

In the first figure, the arcs is not permissible since the region that the blue arc forms with P is not a triangle, while in the second picture, the two arcs τ and τ' do not meet, so they cannot form two sides of a triangle.

Definition 7.4. Two permissible arcs γ, γ' are *equivalent*, written $\gamma \simeq \gamma'$, if

- (a) γ, γ' end in a tile Δ not of type I and $\gamma = \delta_k \cdots \delta_1 \gamma'$, where $\delta_1, \dots, \delta_k$ are consecutive sides of Δ and their first crossing with Δ is with the same side of Δ



- (b) The starting points of γ, γ' in a tile Δ of type I or II , first crossing with P with the same side of Δ , and segments of γ, γ' are isotopic

The equivalence class of a permissible arc γ will be its homotopy class, and will be denoted by $[\gamma]$

- Definition 7.5.**
- (1) Given two curves $\gamma, \gamma' \in (S, M)$, we denote by $I(\gamma, \gamma')$ the number of transversal intersections of representatives of the homotopy classes of γ and γ' . Here, we consider crossings in the interior of S only and not at the boundary. Therefore it is possible that $I(\gamma, \gamma') = 0$, for some arcs γ and γ' . If γ and γ' share an endpoint, and both of the arcs are in P , then $I(\gamma, \gamma') = 0$ always.
 - (2) The intersection vector $I_P(\gamma)$ for some arc γ in (S, M, P) with respect to P is the vector $(I(\gamma, \mathbf{a}_i))_{\mathbf{a}_i \in P}$
 - (3) The intersection number of γ with respect to P is $|I_P(\gamma)| := \sum_{\mathbf{a} \in P} I(\gamma, \mathbf{a})$.
 - (4) A permissible arc γ is trivial if $|I_P(\gamma)| = 0$. In other words, a permissible arc is trivial if it does not intersect any other curve in P

Theorem 7.6. Let P be a partial triangulation of the surface (S, M) . There is a bijection between the equivalence classes of non-trivial permissible arcs in (S, M) and the non-zero strings of A_P . Under

this bijection, the intersection vector corresponds to the dimension vector of the corresponding string module.

Theorem 7.7. Let $S = (S, M)$ be a surface and let P be a partial triangulation of S . Then there is a bijection between the homotopy classes of permissible closed curves c in (S, M) with $|I_P(c)| \geq 2$ and the powers of bands of A_P

8. IRREDUCIBLE MORPHISMS BETWEEN STRING MODULES

Let w be a string and $M(w)$ be the corresponding string module. Then the irreducible morphisms between the string modules are the morphisms obtained by either appending or cutting the string on the left or the right. More specifically they are the morphisms $M(w) \rightarrow M(w_l)$ or $M(w) \rightarrow M(w_r)$, where w_l and w_r are described as follows

- (1) w_l : *Case 1:* There exists an arrow $a \in Q_1$ such that aw is a string. Then let v be the maximal direct string starting at $s(a)$, and then we define w_l to be the string $v^{-1}aw$. This is called adding a hook to $s(w)$, or in other words we are appending the string w with a hook on the left, hence the l in the subscript.

Case 2: There does not exist an arrow $a \in Q_1$. Then in this case, either (i) w is a direct string, or (ii) w can be decomposed as $w = va^{-1}w'$, where v is the maximal direct substring starting at $s(w)$, $a \in Q_1$ and w' is a string. Then here, in (i), $w_l := 0$ and in (ii), $w_l := w'$. This is called removing a cohook from $s(w)$ that is, we are cutting w on the left, and we take the remaining string to be the modified string.

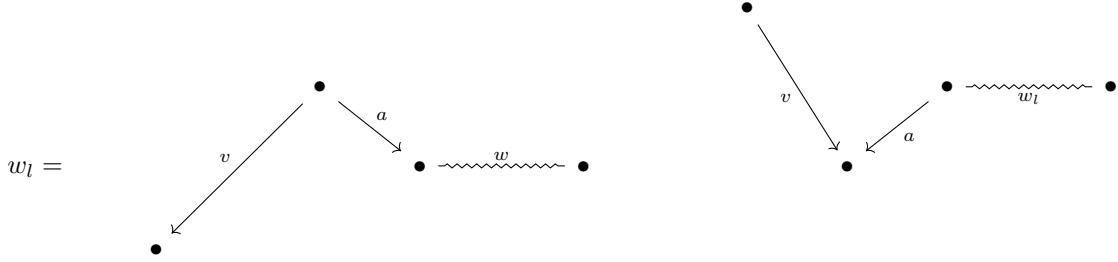


FIGURE 13. Removing and adding a cohook to and from $s(w)$

- (2) w_r : *Case 1:* There is an arrow $a \in Q_1$ such that wa^{-1} is a string. Then let v be the maximal direct string starting at $s(a)$, and we define $w_r := wa^{-1}v$.

Case 2: There is no arrow $a \in Q_1$ such that wa^{-1} is a string. Then in this case, either (i) w is an inverse string, or (ii) w can be decomposed as $w = w'b^{-1}v$, where v is the maximal direct substring starting at $s(w)$, $a \in Q_1$ and w' is a string. Then here, in (i), $w_r := 0$ and in (ii), $w_r := w'$. This is called removing a cohook from w that is, we are cutting w on the right, and we take the remaining string to be the modified string.

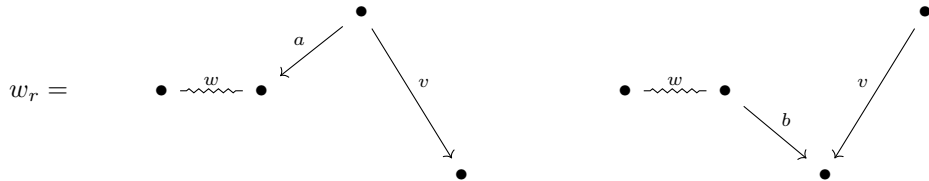
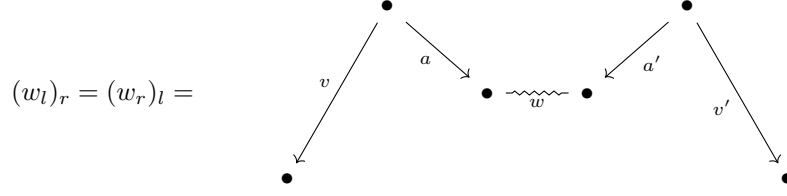
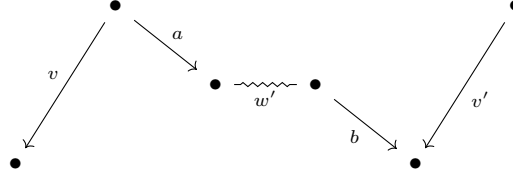
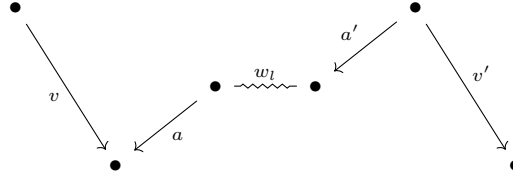


FIGURE 14. Adding a hook on $t(w)$ and removing a cohook from $t(w)$ respectively

Remark 8.1. Clearly, from the following diagrams, we have that $(w_l)_r = (w_r)_l$


 FIGURE 15. Case 1: Adding a hook on the right to $t(w_l)$ or adding a hook on the left to $s(w_r)$

 FIGURE 16. Case 2: Removing a cohook from $t(w_l)$ on the right.

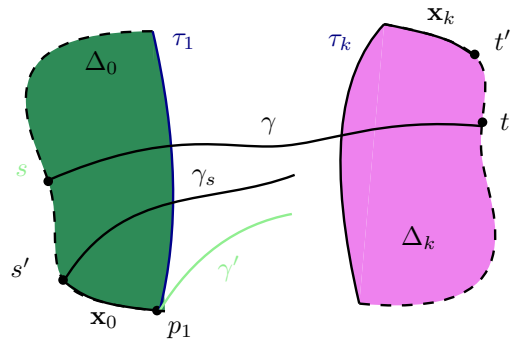
In 16, we decompose $w = w'bv$, where v is a direct string, and we take $(w_l)_r$ to be $v^{-1}aw'$. In other words, we are removing a co-hook from $t(w_l)$


 FIGURE 17. Adding a hook on the right of $t(w)$, the removed cohook of $s(w)$

Proposition 8.2. [3, 2, 1][Lemmas in Page 168 and 169 in [2]] Let w be a string of a gentle algebra and $M(w)$ be the corresponding string module. Then there are at most two irreducible morphisms starting at $M(w)$, namely $M(w) \rightarrow M(w_l)$ or $M(w) \rightarrow M(w_r)$, and these maps are either the natural inclusion or natural projection respectively

9. PIVOT ELEMENTARY MOVES OF ARCS

The question now is how do we interpret these irreducible morphisms on the surface? These morphisms are given by pivot elementary moves of the arcs γ .



Definition 9.1. Let γ be a permissible arc and $M([\gamma])$ denote the corresponding string module with non-empty string. Let s and t be the starting and ending points of γ respectively, and let Δ denote the tile which contains the segment of γ between s and the first crossing with P . Then the pivot elementary move $f_t([\gamma])$ of γ is defined to be the equivalence class of another permissible arc $f_t(\gamma_s)$, where f_t and

γ_s are obtained from γ as follows:

First, we fix t (hence the subscript t in f_t) and move s as according to the following cases:

Case 1: Δ is not of type I or II

We move from s to the vertex s' along the tile Δ counter-clockwise for which we get another arc $\gamma_s \simeq \gamma$, whose endpoints are s' and t and $\gamma \not\simeq \gamma'$, where γ' is defined to be $\gamma_s \circ \mathbf{x}_0$, where $\mathbf{x}_0$ is the side which connects s' and the vertex which is counter clockwise to s' . However, this case splits up into two more cases, one where $\mathbf{x}_0$ is not a boundary segment, and one where $\mathbf{x}_0$ is a boundary segment.

Case 1.1: $\mathbf{x}_0$ is not a boundary segment: In this case, we take $f_t(\gamma_s)$ to be $\gamma_s \circ e$, where e is the boundary segment which connects s' and another vertex counter-clockwise to it. In terms of hooks and cohooks, this adds a hook to $w(\gamma)$

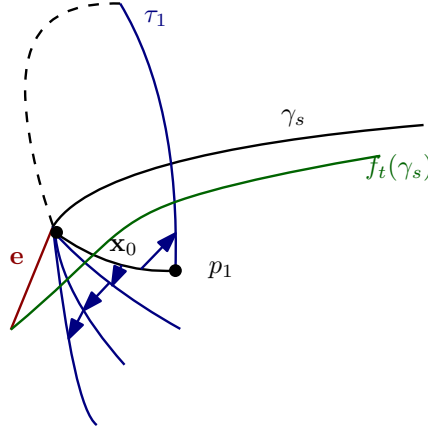


FIGURE 18. $\mathbf{x}_0$ is not a boundary segment

Clearly, from the above figure, a hook is added on the left of $w(\gamma_s)$

Case 1.2: $\mathbf{x}_0$ is a boundary segment: Again here, we take $f_t(\gamma_s) = \gamma_s \circ e$, but in this case, this removes a cohook as from the following figure:

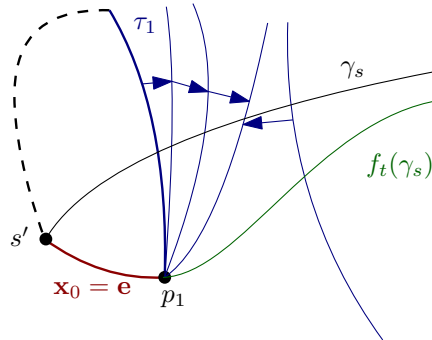
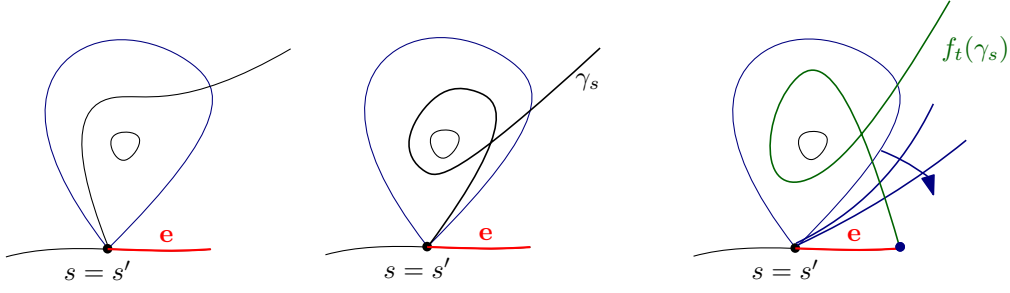
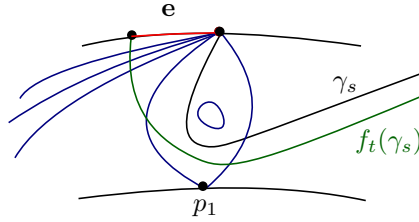


FIGURE 19. $\mathbf{x}_0$ is a boundary segment

Case 2: Δ is of type I. Let γ_s be the permissible arc equivalent to γ , with starting point $s = s'$, such that γ goes around the unmarked boundary component in Δ in the counter clockwise direction. Similarly here, $f_t(\gamma_s) = \gamma_s \circ e$, where e is again the boundary segment connecting s' and the vertex next to it in the counter-clockwise speaking.


 FIGURE 20. Δ is of type I

Case 3: Δ is of type II: Let γ_s be the permissible arc equivalent to γ such that the starting point of γ_s is one of the two marked points in the tile Δ , and the winding number of γ_s around the unmarked boundary component in Δ is zero. Again, $f_t(\gamma_s) = \gamma_s \circ e$, where e is again the boundary segment connecting s' and the vertex next to it in the counter-clockwise sense.


 FIGURE 21. Δ is of type II

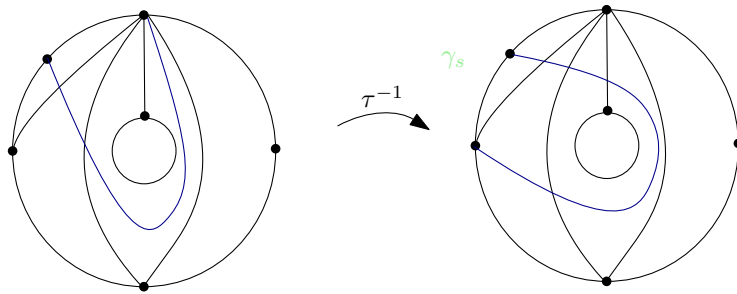
γ_t is obtained by fixing s and moving t , and thus we get that $(\gamma_s)_t = (\gamma_t)_s$, and we denote both by $\gamma_{s,t}$.

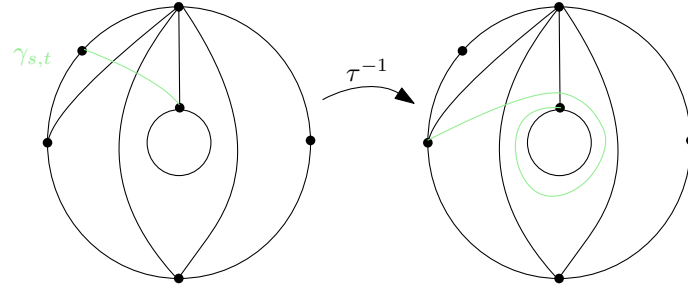
Notation: Let γ be a permissible arc. Then by $\tau^{-1}([\gamma])$, we mean the equivalence class of the arc $\tau^{-1}(\gamma_{s,t})$, which is an arc we obtain by rotating both endpoints to the next point on the boundary counter-clockwise sense.

Corollary: Let $M([\gamma])$ be a string module. We have that $M([\gamma])$ is non-injective if and only if $|I_P(\tau^{-1}(\gamma_{s,t}))| \neq 0$. Moreover, in this case, we have

$$\tau^{-1}(M([\gamma])) = M(\tau^{-1}([\gamma])).$$

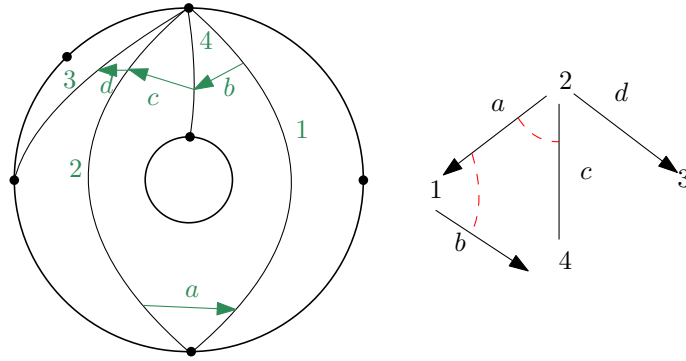
The above corollary means that we can get the Auslander-Reiten translate of a given module simply by taking $\tau^{-1}([\gamma])$ of the given permissible arc


 FIGURE 22. $\tau^{-1}(\gamma_s)$

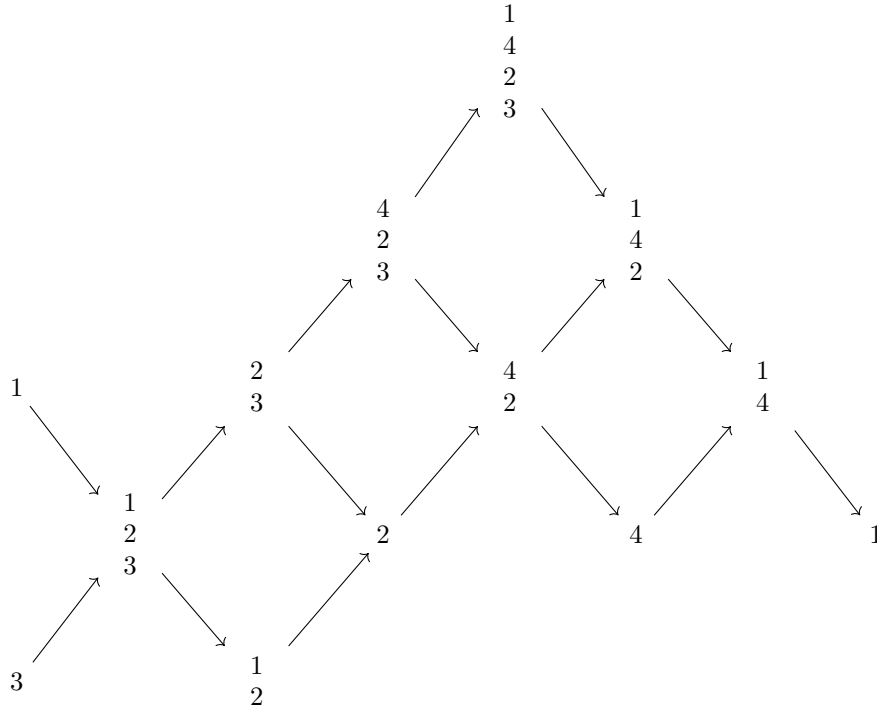
FIGURE 23. $\tau^{-1}(\gamma_{s,t})$

It is important to take $\gamma_{s,t}$ for τ^{-1} , since the images under τ^{-1} might not be the same and figures 21 and 22 illustrate this. The following is an example obtaining the Auslander Reiten quiver of a given triangulation using this geometric model

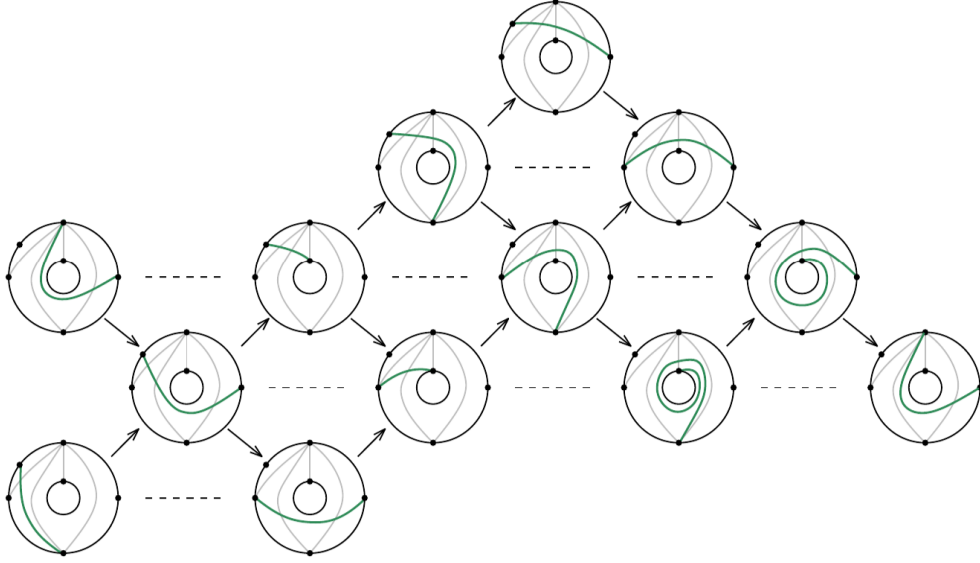
Let us take the following triangulation and corresponding gentle algebra



The Auslander Reiten quiver of this algebra is the following:

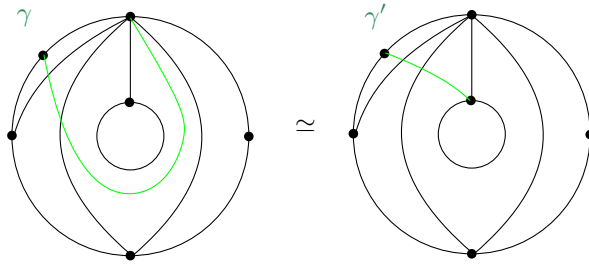


Using this geometric model, the AR quiver can be described in terms of arcs as follows, as from [1] Fig 23,



In this quiver, each equivalence class $[\gamma]$ is represented by the arc $\gamma_{s,t}$, which is the arc that has already been moved, and so the AR translate and τ -orbits are thus taken care of.

Also, in some of the cases, the τ -images of the arcs still follows the "move both endpoints counter-clockwise", but we have replaced them by another representative in the equivalence class. For example, $\tau^{-1}1 = \frac{2}{3}$, and so $\tau^{-1}(M(w(\gamma))) \cong \tau^{-1}(M(w(\gamma')))$, as in the following figure



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